Spring 2021

Marmara University
Faculty of Engineering
Electrical and Electronics Engineering
STAT 2056 - Probability and Random Variables
Midterm Exam Solutions

1. (a) If X is 4, Y can be 4, 3, 2, 1, 0. (4)
If X is -4, Y can be -4, -3, -2, -1, 0 (4)
Hence $S_Y = \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$ (2)

(b) "Transmitted Signal was definetly 4" $= \{(X=4, Y=4), (X=4, Y=3), (X=4, Y=2), (X=4, Y=1)\}$ (10)

(c) $\{(Y=0)\} = \{(X=-4, Y=0), (X=4, Y=0)\}$ (3)
In other words, observed output is zero, we can not determine the input. (2)

2. (a) $P[\text{A car pays } k \text{ TL}] = \left(\frac{1}{2}\right)^k, \quad k = 1, 2, 3, \ldots$ (10)

(b) $P[\text{Car pays 1 TL}] = \frac{1}{2}$ (4)
$P[\text{Car pays 2 TL}] = \frac{1}{4}$ (2)
$P[\text{Car pays 3 TL}] = \frac{1}{8}$ (2)
$P[\text{Car pays 4 TL}] = \frac{1}{16}$ (2)
$P[\text{Car pays 5 TL}] = \frac{1}{32}$ (2)

$P[\text{car pays 6 TL}] = 1 - \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32}\right)$
$= 1 - \left(\frac{16+8+4+2+1}{32}\right)$ (2)
$= 1 - \frac{31}{32} = \frac{1}{32}$

$P[\text{car pays more than 6 TL}] = 0$ (3)

3. (a) Since the number of items is large and the probability is small we can use Poisson approximation.

$\lambda = (\text{items}) \times (\text{probability}) = (10^3 \text{ drives}) \cdot \left(10^{-4} \frac{1}{\text{day}}\right) = 0.1 \frac{\text{drives}}{\text{day}}$ (3)

Let X denote the number of failures in each day.

$P[X=k] = \frac{\lambda^k}{k!} e^{-\lambda} = \frac{(0.1)^k}{k!} e^{-0.1}, \quad k = 0, 1, 2, \ldots$ (2)

(5) $P[X=0] = \frac{(0.1)^0}{0!} e^{-0.1} = e^{-0.1} =$ (5)

(b) In 2 days, on the average we expect $2 \times 0.1 = 0.2$ failures.
Let X denote the number of failures in two days

$P[X=k] = \frac{\lambda^k}{k!} e^{-\lambda} = \frac{(0.2)^k}{k!} e^{-0.2}$ (5) $k = 0, 1, 2, \ldots$ (5)

$P[X<2] = P[X=0] + P[X=1] = \frac{(0.2)^0}{0!} e^{-0.2} + \frac{(0.2)^1}{1!} e^{-0.2} = e^{-0.2} + 0.2 \times e^{-0.2} =$ (4)

4. (a) $E[X] = \frac{1}{8} \times (-4) + \frac{1}{8} \times (-3) + \frac{1}{8} \times (-2) + \frac{1}{8} \times (-1) + \frac{1}{8} \times 0 + \frac{1}{8} \times 1 + \frac{1}{8} \times 2 + \frac{1}{8} \times 3 = -\frac{4}{8} = -\frac{1}{2} = -0.5$

$E[X^2] = \frac{1}{8} \times [(-4)^2 + (-3)^2 + (-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2 + 3^2] = \frac{44}{8} = \frac{11}{2} = 5.5$ (4)

$\text{var}(X) = E[X^2] - (E[X])^2 = \frac{11}{2} - \left(-\frac{1}{2}\right)^2 = \frac{11}{2} - \frac{1}{4} = \frac{22-1}{4} = \frac{21}{4} = 5.25$ (2)

(b) $E[Y] = E[-X^2 + 6] = -E[X^2] + 6 = -\frac{11}{2} + 6 = \frac{-11+12}{2} = \frac{1}{2} = 0.5$ (5)

(3) $E[Y^2] = E[(-X^2+6)^2] = E[X^4 - 12X^2 + 36] = E[X^4] - 12E[X^2] + 36 = E[X^4] - 12 \cdot \frac{11}{2} + 36$

(2) $E[X^4] = \frac{1}{8}[(-4)^4 + (-3)^4 + (-2)^4 + (-1)^4 + 0^4 + 1^4 + 2^4 + 3^4] = \frac{452}{8} = \frac{113}{2} = 56.5$

(2) $E[Y^2] = \frac{113}{2} - 66 + 36 = \frac{113}{2} - 30 = \frac{53}{2} = 26.5$

(3) $\text{var}(Y) = E[Y^2] - (E[Y])^2 = \frac{53}{2} - \left(\frac{1}{2}\right)^2 = \frac{53}{2} - \frac{1}{4} = \frac{106-1}{4} = \frac{105}{4} = 26.25$